

PART 1 – PROJECTILE MOTION

Identification

Directions: Identify the term, concept, variable, or principle described in each statement. Write your answer on the blank.

1. An object that is thrown, kicked, or launched and is acted upon mainly by gravity after release.
Answer: _____
2. The curved path followed by a projectile during its motion.
Answer: _____
3. The two independent motions that combine to produce projectile motion.
Answer: _____
4. The launch angle that produces the maximum horizontal range on level ground.
Answer: _____
5. The launch angle that produces the greatest maximum height.
Answer: _____
6. If a ball is thrown horizontally while another is dropped from the same height at the same time, what happens to both balls?
Answer: _____
7. The horizontal distance traveled by a projectile from launch to landing.
Answer: _____
8. The greatest vertical distance a projectile reaches above its launch point.
Answer: _____
9. If the launch angle remains constant, what happens to the range and maximum height as launch speed increases?
Answer: _____
10. Ignoring air resistance, what is the only force acting on a projectile after it is released?
Answer: _____
11. Two launch angles that add up to 90° and produce the same range are called what?
Answer: _____
12. Between launch angles of 30° and 60° with the same launch speed, which angle produces the greater maximum height?
Answer: _____
13. Why do basketball players often use a high arc when shooting?
Answer: _____
14. Between launch angles of 20° and 70° with the same launch speed, which angle has the shorter flight time?
Answer: _____
15. How does the horizontal velocity of a projectile behave during flight when air resistance is ignored?
Answer: _____
16. What is the vertical velocity of a projectile at the highest point of its trajectory?
Answer: _____

17. Compared with the ideal prediction, what effect does air resistance have on a projectile's range?
Answer: _____
18. What launch condition helps a long jumper achieve maximum range?
Answer: _____
19. If the launch speed doubles while the launch angle remains at 45° , approximately how many times greater will the range become?
Answer: _____
20. Which two variables determine the range and maximum height of a projectile?
Answer: _____
21. At what point in its flight is a projectile's total speed the slowest?
Answer: _____
22. At what positions along the trajectory does a projectile have equal speed due to symmetry?
Answer: _____
23. Why do archers aim slightly above the target?
Answer: _____
24. Which pair of launch angles produces the same range but different maximum heights?
Answer: _____
25. Does horizontal motion affect the time required for vertical free fall?
Answer: _____
26. As the launch angle increases from 0° to 45° to 90° , how do the range and maximum height change?
Answer: _____
27. Why does a launch angle of 45° produce the maximum range on level ground?
Answer: _____
28. In an experiment with a fixed launch angle, how does increasing launch speed affect the range?
Answer: _____
29. Which of the following is **not** an example of projectile motion?
Answer: _____
30. What is the shape of the ideal path followed by a projectile?
Answer: _____
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Answer Key

1. Projectile
2. Trajectory
3. Constant horizontal motion and accelerated vertical free fall
4. 45°
5. 90° (straight upward)
6. Both hit the ground at the same time
7. Range

8. Maximum height
9. Both range and height increase
10. Gravity
11. Complementary angles
12. 60°
13. More flight time and a better entry angle to the hoop
14. 20°
15. Remains constant
16. Zero
17. Makes the range shorter
18. Fast run-up with a launch angle near 45°
19. Four times
20. Launch speed and launch angle
21. At the peak (top of the trajectory)
22. Any two points at the same height above the launch point
23. To compensate for gravity pulling the arrow downward
24. Any pair of complementary angles (e.g., 40° and 50°)
25. No
26. Range peaks at 45° then decreases; maximum height continues to increase
27. It balances horizontal speed and flight time
28. It increases approximately as the square of the launch speed
29. Driving a car on a straight, flat road
30. Parabola

PART 2 – MOMENTUM AND COLLISIONS

Identification

Directions: Identify the term, concept, formula, collision type, or principle described in each statement. Write your answer on the blank.

1. The quantity defined as the product of an object's mass and velocity.
Answer: _____
2. The formula used to calculate momentum.
Answer: _____
3. Momentum describes how difficult it is to _____ a moving object.
Answer: _____
4. The two factors that determine an object's momentum.
Answer: _____
5. A collision in which no kinetic energy is lost and the objects bounce apart.
Answer: _____

6. A collision in which the objects stick together or deform, causing kinetic energy to decrease.
Answer: _____
7. In an isolated system, the total momentum before a collision is equal to the total momentum _____.
Answer: _____
8. The product of force and the time during which it acts.
Answer: _____
9. Seatbelts and airbags reduce the force during a collision by increasing the _____.
Answer: _____
10. For objects with equal mass, what happens to momentum as speed increases?
Answer: _____
11. Billiard balls colliding and bouncing off with nearly the same speed are an example of what type of collision?
Answer: _____
12. Train cars coupling together after a collision demonstrate what type of collision?
Answer: _____
13. Between a jeepney and a motorcycle traveling at the same speed, which has the greater momentum?
Answer: _____
14. Catching an egg by moving your hand backward increases the _____, reducing the force on the egg.
Answer: _____
15. The recoil of a cannon after firing is explained by what physical principle?
Answer: _____
16. Crumple zones in vehicles reduce the force of impact by increasing the _____ and _____.
Answer: _____
17. In any collision involving an isolated system, what quantity is always conserved?
Answer: _____
18. The form of energy that is conserved only in elastic collisions.
Answer: _____
19. A 1 kg object moving at 5 m/s sticks to a 4 kg object at rest. What is their final speed?
Answer: _____
20. Why is falling onto a mattress less painful than falling onto concrete?
Answer: _____
21. Momentum is the product of what two quantities?
Answer: _____
22. Two balls moving at equal speeds toward each other and bouncing apart demonstrate what type of collision?
Answer: _____
23. A boxer's punch has large momentum mainly because of the _____ of the fist.
Answer: _____
24. During a car crash, what effect does a longer impact time have on the force experienced?
Answer: _____

25. The recoil of a fire hose when water is released demonstrates what law?

Answer: _____

26. According to the impulse-momentum theorem, the product of force and time equals the _____.

Answer: _____

27. What is the main difference between elastic and inelastic collisions?

Answer: _____

28. When momentum is conserved during a collision, what quantity is exchanged between the objects?

Answer: _____

29. Which object is harder to stop: a heavy, slow-moving object or a light, fast-moving object?

Answer: _____

30. Safety devices such as airbags and seatbelts reduce injuries by increasing _____ to decrease force.

Answer: _____

Answer Key

1. Momentum
2. $\mathbf{p = m \times v}$
3. Stop
4. Mass and velocity
5. Elastic collision
6. Inelastic collision
7. After the collision
8. Impulse
9. Stopping time (impact time)
10. Momentum increases
11. Nearly elastic collision
12. Perfectly inelastic collision
13. Jeepney
14. Impact time (stopping time)
15. Conservation of momentum
16. Stopping distance and stopping time
17. Total momentum
18. Kinetic energy
19. **1 m/s**
20. Longer stopping time results in a smaller force
21. Mass and velocity
22. Elastic collision
23. High velocity
24. It reduces the force

25. Conservation of momentum
26. Change in momentum (Δp)
27. Kinetic energy is conserved in elastic collisions but not in inelastic collisions; momentum is conserved in both
28. Momentum
29. It depends on the product of mass and velocity (momentum)
30. Stopping time

PART 4 – RENEWABLE AND NONRENEWABLE ENERGY

Identification

Directions: Identify the term, concept, energy source, or principle described in each statement. Write your answer on the blank.

1. Energy sources that are depleted after use and require millions of years to form.
Answer: _____
2. Energy sources that are naturally replenished within a relatively short period of time.
Answer: _____
3. The primary source of electricity generation in the Philippines today.
Answer: _____
4. The renewable energy resource in which the Philippines ranks second in the world.
Answer: _____
5. The greenhouse gas mainly released when fossil fuels are burned.
Answer: _____
6. Energy produced from the movement or falling of water.
Answer: _____
7. Energy obtained directly from sunlight.
Answer: _____
8. Energy produced from organic materials such as agricultural waste, wood, and crops.
Answer: _____
9. The major nonrenewable natural gas source in the Philippines.
Answer: _____
10. Why do renewable energy sources help reduce climate change?
Answer: _____
11. Identify the group composed only of nonrenewable energy sources.
Answer: _____
12. Identify the group composed only of renewable energy sources commonly used in the Philippines.
Answer: _____

13. What is the major environmental effect of coal-fired power plants?
Answer: _____
 14. What is the primary disadvantage of constructing large hydroelectric dams?
Answer: _____
 15. Why is geothermal energy highly suitable for the Philippines?
Answer: _____
 16. What economic benefit results from increasing the use of renewable energy sources?
Answer: _____
 17. Solar and wind energy are classified as what type of energy sources because their output depends on weather conditions?
Answer: _____
 18. Why is natural gas considered a transition fuel?
Answer: _____
 19. Replacing coal-fired power plants helps reduce what health problem?
Answer: _____
 20. State one major advantage of wind energy.
Answer: _____
 21. Compare the long-term cost of renewable energy with fossil fuels.
Answer: _____
 22. The Philippine goal of achieving 35% renewable energy by 2030 and 50% by 2040 primarily supports what two objectives?
Answer: _____
 23. Geothermal and hydropower are classified as what type of renewable energy because they can provide continuous electricity?
Answer: _____
 24. What social issue is commonly associated with large energy development projects?
Answer: _____
 25. Besides fuel costs, what other impacts should be included in the true cost of energy production?
Answer: _____
 26. Coal ash and mining waste may release what harmful materials into the environment?
Answer: _____
 27. Energy choices affect what four important aspects of society and electricity use?
Answer: _____
 28. Give a reliable clean energy mix recommended for the Philippines.
Answer: _____
 29. Why are renewable energy prices generally more stable than fossil fuel prices?
Answer: _____
 30. Why is learning about renewable and nonrenewable energy important for students?
Answer: _____
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Answer K

1. Nonrenewable energy
2. Renewable energy
3. Coal
4. Geothermal energy
5. Carbon dioxide (CO₂)
6. Hydropower (Hydroelectric power)
7. Solar energy
8. Biomass energy
9. Natural gas (Malampaya)
10. They emit little to no CO₂ during operation
11. Coal, oil, and natural gas
12. Geothermal, hydropower, solar, wind, and biomass
13. Air pollution and high carbon dioxide emissions
14. Flooding and displacement of communities
15. The Philippines is located along the Ring of Fire
16. Lower fuel imports and greater price stability
17. Intermittent energy sources
18. It is cleaner than coal but is still a finite fossil fuel
19. Respiratory illnesses caused by air pollution
20. Free fuel and low emissions
21. Renewable energy has higher installation costs but lower long-term operating costs
22. Climate goals and energy security
23. Baseload renewable energy sources
24. Local communities often bear the negative impacts of the projects
25. Health, environmental, and climate impacts
26. Toxic substances
27. Cost, reliability, emissions, and electricity bills
28. Geothermal + Hydropower + Solar + Wind + Energy Storage
29. They have no fuel cost once the facilities are built
30. Because today's energy choices will affect people's lives and the environment in the future.